

SPECS

Supermarket Pricing Estimation & Comparison System

Bishop Stuart University · Mbarara City, Uganda
Department of Computing, Library, Information and Technology
Diploma in Information Technology — Internet and Web Programming

Registration Nos: 24/BSU/DIT/3253, 24/BSU/DCS/1450, 24/BSU/DIT/1554,
24/BSU/DIT/1423, 24/BSU/DIT/2916, 24/BSU/DCS/0758
<https://github.com/ALVINeald/SPECS-GROUP-6>

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1. About the Project

SPECS is a web-based platform that enables consumers in Mbarara City to compare supermarket prices across multiple stores, build smart shopping baskets, set price drop alerts, and generate optimised multi-stop shopping routes, all in one place.

The platform covers 7 supermarkets, 205+ products, and 1,385+ prices, updated by administrators and managers in real time.

2. Covered Supermarkets

Store	Tier	Location
FRESCO Supermarket	Premium	Buremba Road
Kirimi Supermarket	Mid	Buremba Road
Day to Day	Budget	High Street
Apple Door to Door	Mid	Bananuka Drive
Amazon Express	Budget	High Street
Golf Course Supermarket	Premium	Lower Circular Road
Mbarara Central Market	Market	Buremba Road

3. Key Features

- Price Comparison: Browse 205+ products and instantly see prices across all 7 stores side by side.
 - Smart Basket: Add items and split your shopping across stores for maximum savings
 - Shopping Route: Interactive Leaflet.js map showing the optimal multi-stop route
 - Price Alerts: Set a target price and get notified when a product drops below your budget
 - Price Trends: Chart.js graphs showing how prices change over time
 - Admin Panel: Full product, price, store, and user management with audit logs
 - Reports: Price summary, store comparison, price changes, and user activity reports with CSV export
 - Forgot Password: Secure token-based password reset flow
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4. Tech Stack

Layer	Technology
Backend	PHP 8.2
Database	MySQL / MariaDB
Local Server	XAMPP
Frontend	HTML5, CSS3, Vanilla JavaScript
Maps	Leaflet.js
Charts	Chart.js
DB Admin	phpMyAdmin / HeidiSQL
Version Control	Git / GitHub

5. Database Overview

Database name: specs_db

Item	Count
Tables	11
Views	3
Stored Procedures	2
Triggers	1
Products	205
Stores	7
Prices	1,385+

Tables: categories, stores, users, products, prices, price_history, alerts, basket, store_plans, admin_logs, password_resets

6. Installation & Setup

Requirements

- XAMPP (PHP 8.x + MySQL)
- Web browser

Steps

1. Copy the project folder into C:\xampp\htdocs\specs\
2. Start XAMPP — ensure Apache and MySQL are running
3. Open phpMyAdmin at <http://localhost/phpmyadmin>
 - Create a new database called specs_db
 - Import specs_db.sql (structure)
 - Import specs_seed_v2.sql (data)
4. Visit http://localhost/specs/generate_passwords.php then delete the file
5. Open the app at <http://localhost/specs/login.php>

7. Default Accounts

Name	Email	Password	Role
Mbabazi Alvin	alvin@specs.ug	specsmbarara2025	Admin
SPECS Administrator	admin@specs.ug	admin123	Admin
Store Manager	manager@specs.ug	manager123	Manager
Sarah Tumwebaze	sarah@gmail.com	user123	User
John Muhwezi	john@yahoo.com	user123	User
Grace Atuhairu	grace@gmail.com	user123	User

⚠ **Change all passwords before any public deployment.**

8. Design System

Token	Value	Usage
--forest	#18382a	Primary dark green
--leaf	#2d6a4f	Buttons, headings
--mint	#52b788	Accents, highlights
--gold	#e9a820	Alerts, badges
--cream	#fdf8f2	Page backgrounds
--sand	#e8e2d9	Borders, dividers
--ink	#1c1a17	Body text
--muted	#7a7060	Secondary text

Fonts: Nunito (headings) + Nunito Sans (body)

9. Security Features

- Passwords hashed with bcrypt (cost = 12)
 - Prepared statements throughout — SQL injection prevention
 - Session-based authentication with role checks
 - Token-based forgot password flow
 - Admin audit logs for all sensitive actions
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10. Author

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